1 INTRODUCTION

1.1 Background of the Study

In recent years, e-commerce platforms have become one of the most important channels for buying and selling products online. Platforms such as Amazon, Flipkart, Myntra, and Meesho allow customers to browse products, compare prices, read reviews, and make purchases conveniently from their mobile devices or computers.

One of the most common behaviors observed in online shopping is the **shopping cart feature**. Customers can add products to their shopping cart before completing the checkout process. However, a large number of customers leave the platform without completing the purchase after adding items to the cart. This behavior is known as **cart abandonment**.

Cart abandonment has become a major concern for e-commerce businesses. Studies show that a significant percentage of online shoppers abandon their carts before completing the purchase. This leads to lost revenue opportunities and affects the overall performance of online businesses.

Understanding why customers abandon their carts is extremely important for improving the online shopping experience. Various factors influence cart abandonment, including high product prices, additional delivery charges, complicated checkout procedures, payment difficulties, slow delivery times, and technical issues on the platform.

With the help of **data analytics and visualization techniques**, businesses can analyze customer behavior and identify patterns related to cart abandonment. By analyzing customer data, organizations can understand which factors influence purchasing decisions and take appropriate actions to reduce cart abandonment.

This project focuses on analyzing customer responses related to online shopping behavior and identifying the major reasons behind cart abandonment. The data used for this project was collected through a **Google Form survey** shared with online shoppers.

The collected dataset includes information about customer demographics, online shopping frequency, preferred shopping platforms, purchase completion status, and the reasons why customers abandon their carts.

The dataset was cleaned and analyzed using **Python libraries such as Pandas and NumPy**, and the results were visualized using **Power BI dashboards**. These visualizations help in identifying important patterns in customer behavior and provide insights that can help e-commerce platforms improve their services.

By understanding customer behavior and identifying the key reasons for cart abandonment, businesses can develop strategies to improve the shopping experience, increase customer satisfaction, and ultimately increase sales conversion rates.

1.2 Problem Statement

Cart abandonment is one of the major challenges faced by e-commerce platforms. Customers frequently add products to their shopping carts but leave the platform without completing the purchase process.

This behavior leads to several problems for businesses, including loss of potential revenue, increased marketing costs, and reduced conversion rates. Companies invest significant resources in attracting customers to their websites through advertisements and promotions. However, when customers abandon their carts without completing the purchase, these efforts may not result in actual sales.

The reasons behind cart abandonment may vary from customer to customer. Some customers abandon their carts due to high product prices, additional delivery charges, or long delivery times. Others may face technical issues during the checkout process or experience difficulties with payment methods.

In many cases, e-commerce platforms do not fully understand the specific reasons behind customer abandonment behavior. Without proper analysis, it becomes difficult for businesses to identify the main problems affecting their sales performance.

Therefore, there is a need for a **data-driven system that can analyze customer shopping behavior, identify the key reasons for cart abandonment, and provide meaningful insights to help businesses reduce abandonment rates.**

1.3 Objectives

The main objectives of this project are:

Objective 1

To analyze customer behavior in online shopping platforms and understand how customers interact with e-commerce applications. This includes studying customer demographics such as age group and gender, as well as shopping habits such as online shopping frequency and preferred shopping platforms.

Objective 2

To identify the major reasons behind cart abandonment and visualize these insights using data analytics and visualization tools such as Python and Power BI.

1.4 Scope of the Project

The scope of this project is focused on analyzing customer behavior related to online shopping and cart abandonment.

The system analyzes customer responses collected through a survey and identifies patterns in customer shopping behavior. The analysis helps understand which factors influence customer decisions during the purchase process.

The insights obtained from this project can be useful for:

• E-commerce businesses that want to reduce cart abandonment rates
• Online retailers who want to improve the customer checkout experience
• Marketing teams who want to understand customer preferences
• Researchers studying consumer behavior in online shopping

The results of this project can help businesses implement strategies such as offering discounts, reducing delivery charges, simplifying checkout procedures, and improving payment options to encourage customers to complete their purchases.

1.5 Limitations

Although this project provides valuable insights into cart abandonment behavior, it has certain limitations.

• The dataset used in this project was collected through a survey and may not represent the entire population of online shoppers.

• The analysis is based on customer responses, which may vary depending on individual experiences and preferences.

• The revenue loss estimation is based on an assumed average cart value and may not represent actual business revenue.

• The project focuses mainly on customer behavior analysis and does not include real transaction data from e-commerce platforms.

Despite these limitations, the project provides useful insights into the major factors influencing cart abandonment.

1.6 Module Description

The system consists of several modules that work together to analyze customer behavior and identify the reasons for cart abandonment.

1.6.1 Data Collection Module

The first step in the project is collecting customer responses related to online shopping behavior. In this project, data was collected using a **Google Form survey**.

The survey collected information such as:

• Age group of the customer
• Gender
• Online shopping frequency
• Preferred shopping applications
• Purchase completion status
• Reasons for abandoning the shopping cart
• Suggestions for improving the shopping experience

The responses collected through Google Forms were exported into an **Excel dataset**, which was used as the input for further analysis.

1.6.2. Data Preprocessing Module

Raw datasets often contain inconsistencies such as missing values, duplicate entries, or unstructured data. Therefore, data preprocessing is required before performing analysis.

In this module, Python libraries such as **Pandas and NumPy** were used to clean and organize the dataset.

The preprocessing steps include:

• Removing duplicate records
• Handling missing values
• Standardizing text values
• Splitting multiple values stored in a single column
• Converting categorical values into consistent formats

After preprocessing, the dataset becomes clean and structured, making it suitable for analysis.

1.6.3. Exploratory Data Analysis Module

Exploratory Data Analysis (EDA) is performed to understand the structure of the dataset and identify patterns in customer behavior.

In this stage, several analyses were performed, including:

• Distribution of customers across different age groups
• Gender distribution of respondents
• Online shopping frequency analysis
• Purchase completion behavior
• Preferred shopping applications

These analyses help identify trends in customer behavior and understand how different customer groups interact with online shopping platforms.

1.6.4. Risk Score Analysis Module

Risk score analysis is used to categorize customers based on their likelihood of abandoning their shopping carts.

Customers were classified into three categories:

High Risk: Customers who frequently abandon their carts and rarely complete purchases.

Medium Risk: Customers who sometimes complete purchases but occasionally abandon carts.

Low Risk: Customers who usually complete their purchases.

This classification helps businesses identify customers who require targeted marketing strategies to reduce cart abandonment.

1.6.5. Revenue Loss Analysis Module

Revenue loss analysis is performed to estimate the potential financial loss caused by cart abandonment.

In this project, the revenue loss was calculated using the following approach:

- Count the number of customers who abandoned their carts.
- Assume an average cart value.
- Multiply the number of abandoned carts by the average cart value.

For example, if the average cart value is **₹1000** and **62 customers abandon their carts**, the estimated revenue loss would be:

Revenue Loss = 62 × 1000 = **₹62,000**

This analysis helps businesses understand the financial impact of cart abandonment.

1.6.6. Data Visualization Module

The final step of the implementation process involves visualizing the analysis results using **Power BI dashboards**.

The dashboard includes several charts that represent important insights about customer behavior.

The visualizations include:

• Age group distribution chart
• Gender distribution chart
• Online shopping frequency chart
• Purchase status analysis
• Risk level classification chart
• Preferred shopping applications chart
• Cart abandonment reasons analysis
• Revenue loss analysis
• Potential revenue recovery visualization

These visualizations help businesses easily interpret the data and make informed decisions to improve the online shopping experience.

2 SYSTEM ANALYSIS

2.1 Existing System

In the existing e-commerce environment, online shopping platforms allow customers to browse products, add items to their shopping carts, and complete purchases through an online checkout process. Most platforms track basic customer activities such as product views, cart additions, and completed transactions. However, the analysis of customer behavior related to cart abandonment is often limited.

In many cases, businesses focus mainly on sales performance without deeply analyzing the reasons why customers abandon their shopping carts. When a customer adds a product to the cart but does not complete the purchase, the platform may only send reminder notifications or promotional emails to encourage the customer to complete the purchase.

Process in Existing System

The typical process followed in the existing system includes:

- Customer visits an e-commerce website or mobile application.
- Customer browses products and adds items to the shopping cart.
- Customer proceeds to the checkout page.
- If the customer does not complete the payment, the cart is abandoned.
- The system records the abandoned cart but often does not analyze the detailed reasons behind the abandonment.

Problems in Existing System

The existing system has several limitations that affect business performance.

• Limited analysis of customer behavior patterns.
• Lack of detailed understanding of cart abandonment reasons.
• Businesses often rely on general assumptions instead of data-driven insights.
• No structured analysis of demographic factors such as age and gender.
• Difficulty in identifying high-risk customers who frequently abandon carts.
• Limited visualization of customer shopping patterns.

Because of these limitations, businesses may struggle to implement effective strategies to reduce cart abandonment and improve conversion rates.

2.2 Proposed System

The proposed system is a **Cart Abandonment Analysis and Visualization System** that analyzes customer responses related to online shopping behavior and identifies the key reasons behind cart abandonment.

The system uses **data analytics techniques to process and analyze customer survey data**. By applying data preprocessing, exploratory data analysis, and visualization techniques, the system provides meaningful insights into customer shopping behavior.

In this project, customer responses were collected through a **Google Form survey**. The dataset includes important information such as customer demographics, shopping frequency, preferred e-commerce platforms, purchase status, and reasons for cart abandonment.

The collected dataset was processed using **Python libraries such as Pandas and NumPy** to clean and organize the data. Exploratory Data Analysis (EDA) was performed to identify patterns and trends in customer behavior.

The results of the analysis were visualized using **Power BI dashboards**, which provide graphical representations of important insights such as customer demographics, shopping frequency, preferred platforms, cart abandonment reasons, and estimated revenue loss.

Working of the Proposed System

The proposed system follows the following steps:

- Customer responses are collected through a Google Form survey.
- The dataset is exported to an Excel file.
- Data preprocessing is performed using Python to clean and organize the dataset.
- Exploratory Data Analysis is performed to identify patterns in customer behavior.
- Risk score analysis is performed to classify customers based on abandonment likelihood.

- Revenue loss analysis is conducted to estimate the financial impact of cart abandonment.
- The analyzed data is visualized using Power BI dashboards.
- Insights and recommendations are generated to help reduce cart abandonment.

The proposed system provides businesses with a better understanding of customer behavior and helps identify strategies to improve the online shopping experience.

Advantages of Proposed System

The proposed system offers several advantages compared to the existing system.

• Provides detailed analysis of customer behavior in online shopping platforms.
• Identifies the major reasons behind cart abandonment.
• Uses data analytics techniques to generate meaningful insights.
• Visualizes customer behavior using interactive Power BI dashboards.
• Helps businesses estimate potential revenue loss caused by abandoned carts.
• Enables data-driven decision making for improving the shopping experience.
• Helps e-commerce platforms implement strategies to reduce cart abandonment.
• Improves overall sales performance by increasing purchase completion rates.

2.3 System Specification

The system specification defines the technical requirements needed for the successful implementation of the Cart Abandonment Analysis and Visualization System. It includes both hardware and software requirements required to perform data collection, preprocessing, analysis, and visualization.

2.3.1 Software Requirements

The implementation of the **Cart Abandonment Analysis and Visualization System** requires several software tools and technologies for data processing, analysis, and visualization. These tools help transform raw customer survey data into meaningful insights.

Programming Language:

Python 3.x

Development Environment:

Jupyter Notebook

Data Visualization Tool:

Power BI

Python Libraries Used:

Pandas
NumPy
Matplotlib
Seaborn

Data Source

- Google Forms (Survey Data)
- Excel Dataset (.xlsx)

Algorithms Used:

- Rule-Based Algorithm (Risk Score Classification)
- Formula-Based Algorithm (Revenue Loss Estimation)
- Data Processing Techniques (Using Pandas and NumPy)

2.3.2 Hardware Requirements

The system does not require high-end hardware resources since the project mainly focuses on data analysis and visualization.

The minimum hardware requirements for running the system include:

Processor: Intel i3 or above
RAM: Minimum 4 GB
Storage: At least 10 GB free disk space
Operating System: Windows / Linux / macOS

2.3.3 About Software

Programming Language

Python 3.x

Python was used for performing data preprocessing, exploratory data analysis, and revenue loss estimation. Python provides powerful libraries for data manipulation and analysis, making it suitable for handling structured datasets.

Development Environment

Jupyter Notebook

Jupyter Notebook was used as the primary development environment for implementing the data analysis process. It allows interactive coding, visualization, and documentation in a single environment.

Data Visualization Tool

Power BI

Power BI was used to create interactive dashboards that visually represent customer behavior patterns, cart abandonment reasons, and revenue loss analysis.

Python Libraries Used

Several Python libraries were used during the implementation phase.

Pandas

Pandas is used for data manipulation and preprocessing. It helps in reading the dataset, cleaning the data, and performing various transformations on the dataset.

NumPy

NumPy is used for numerical computations and handling arrays efficiently.

Matplotlib

Matplotlib is used to create basic data visualizations such as charts and graphs.

Seaborn

Seaborn is used for advanced statistical visualizations and provides more attractive chart styles.

3 SYSTEM DESIGN

System design is the process of defining the architecture, components, modules, and data flow of a system. It provides a structured framework that explains how the system will function and how different components interact with each other.

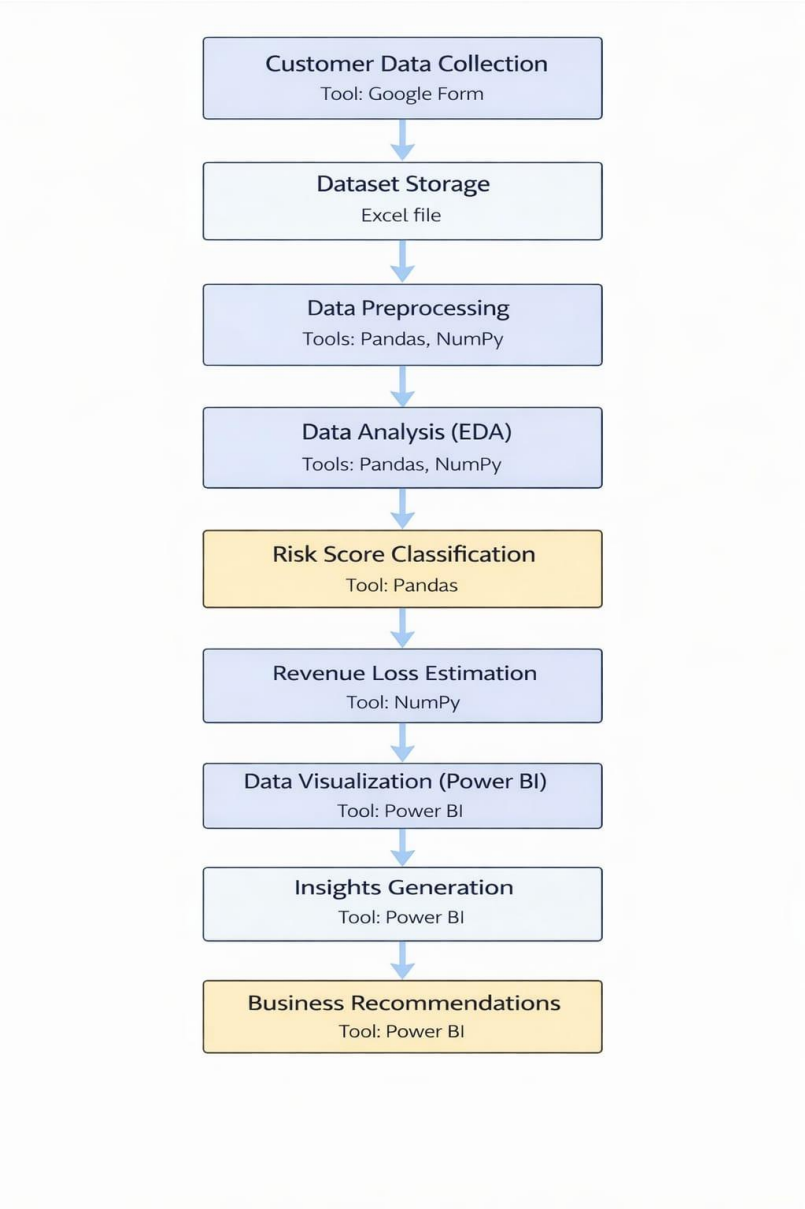
In this project, the system design focuses on developing a **Cart Abandonment Analysis and Visualization System** that processes customer data, analyzes behavior, and generates meaningful insights

3.1 System notation

System notation refers to the set of standard symbols and graphical representations used to describe the structure and functionality of a system. These notations help in visually representing how data flows, how processes are carried out, and how different components of the system interact.

In this project, system notation is used to represent the working of the **Cart Abandonment Analysis and Visualization System** in a clear and understandable manner.

3.1.1 Data Flow Diagram



3.1.3 Use Case Diagram

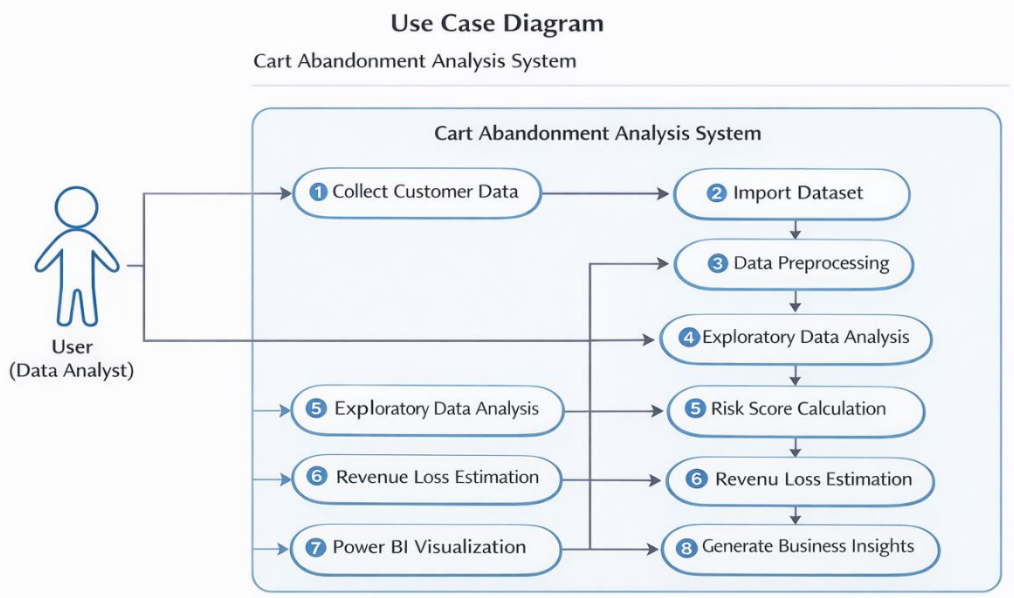


Figure 3.3: Use Case Diagram for Cart Abandonment Analysis System

3.2 Design Types

Design types refer to the different approaches used in system design to organize and structure a system at various levels, such as overall architecture and detailed implementation.

3.2.1 Input Design

Input design is the process of designing how data is collected and entered into the system. It ensures that the input data is accurate, complete, and easy to use. Proper input design helps in reducing errors and improving the quality of data analysis.

In this project, input design focuses on collecting customer data related to online shopping behavior and cart abandonment.

Input Source

The input data for this system is collected through:

- **Google Forms Survey**
- Exported as **Excel Dataset (.xlsx)**

The survey collects responses from users about their shopping behavior.

Input Data Fields

The system accepts the following input fields:

- Name
- Age Group
- Gender
- Online Shopping Frequency
- Preferred Shopping Application
- Added to Cart
- Purchase Status
- Abandonment Reason
- Checkout Importance
- Discount Notification
- Reminder Notification
- Suggestions

3.2.2 Dataset Design

Dataset design refers to the structure and organization of data used in the system. It defines how data is stored, categorized, and represented for analysis.

In this project, the dataset is designed to capture customer behavior related to online shopping and cart abandonment.

Dataset overview:

Attribute	Description	Data Type
Name	Name of the survey respondent	Categorical
Age_Group	Age category of the customer (18–24, 25–34, etc.)	Categorical
Gender	Gender of the customer	Categorical
Online_Shopping_ Frequency	Indicates how frequently the customer shops online	Categorical
Preferred_Shopping_App	The e-commerce platforms preferred by the customer	Categorical
Added_To_Cart	Indicates whether the customer added items to the cart	Categorical
Purchase_Status	Shows whether the customer completed the purchase	Categorical
Reason_Category	Reason why the customer abandoned the cart	Categorical
Checkout_Importance	Importance of the checkout process for customers	Categorical
Discount_Notification	Indicates whether discounts influence purchasing decisions	Categorical

Remainder_Notification	Indicates whether reminders help customers complete purchases	Categorical
Suggestion_1	Customer suggestions for improving shopping experience	Text
Suggestion_2	Additional suggestions provided by customers	Text

3.2.3 Output Design

The output of this project, Cart Abandonment Analysis and Visualization System, is designed to present the analyzed customer data in a clear and understandable format. The results of the analysis, such as age group distribution, gender-wise analysis, cart abandonment reasons, risk levels, and revenue loss, are effectively visualized using Power BI dashboards.
The visualizations include charts and graphs that help in identifying patterns and trends in customer behavior. These outputs make it easier for users to interpret the data and support better decision-making to reduce cart abandonment.

Visualization Analysis (Charts Used)

In this project, different types of charts were used in Power BI to represent various aspects of customer behavior. Each chart was selected based on the type of data and analysis required.

Age Group Analysis

For age group distribution, a **bar chart** was used to compare the number of customers in different age categories.

Gender Analysis

For gender distribution, a **pie chart** was used to represent the proportion of male and female customers.

Online Shopping Frequency Analysis

For analyzing shopping frequency, a **column chart** was used to show how often customers shop online.

Online Shopping Frequency by Age and Gender

For comparing shopping frequency across age groups and gender, a **clustered column chart** was used.

Preferred Shopping Applications

For analyzing preferred platforms, a **bar chart** was used to compare different e-commerce applications.

Purchase Status Analysis

For purchase behavior, a **column chart** was used to compare completed and abandoned purchases.

Age Group vs Purchase Status

For analyzing purchase behavior across age groups, a **stacked column chart** was used.

Cart Abandonment Reasons

For identifying reasons for cart abandonment, a **horizontal bar chart** was used for better readability.

Risk Level Analysis

For risk classification, a **pie chart** was used to show the distribution of high, medium, and low risk customers.

Revenue Loss Analysis

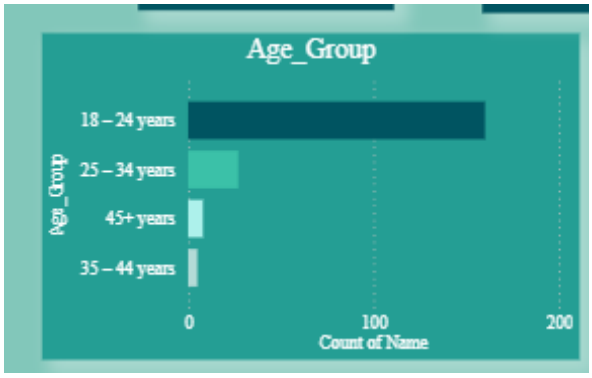
For estimating revenue loss, a **bar chart** was used to compare losses across different factors.

Revenue Recovery Analysis

For showing potential recovery, a **column chart** was used to represent different improvement levels.

4 RESULTS AND INSIGHTS

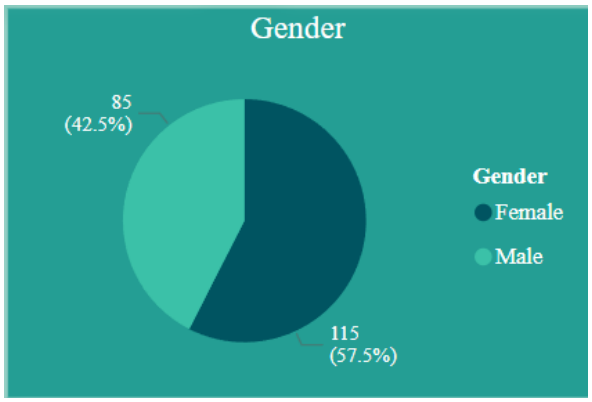
Age Group Analysis



Insight:

Most respondents belong to the **18–24 age group**, showing that younger users are the most active in online shopping.

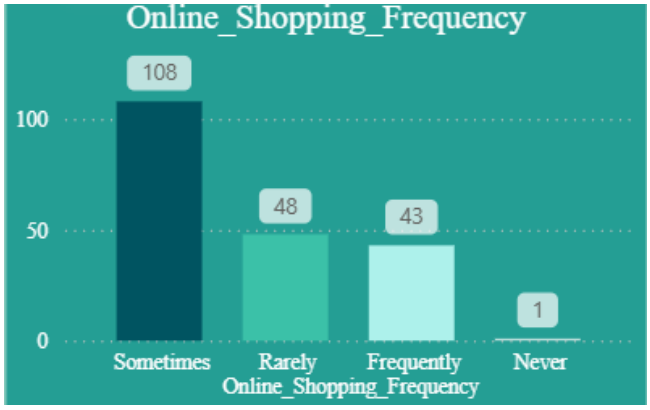
Gender Distribution



Insight:

Female customers slightly outnumber male customers, indicating strong participation of women in online shopping.

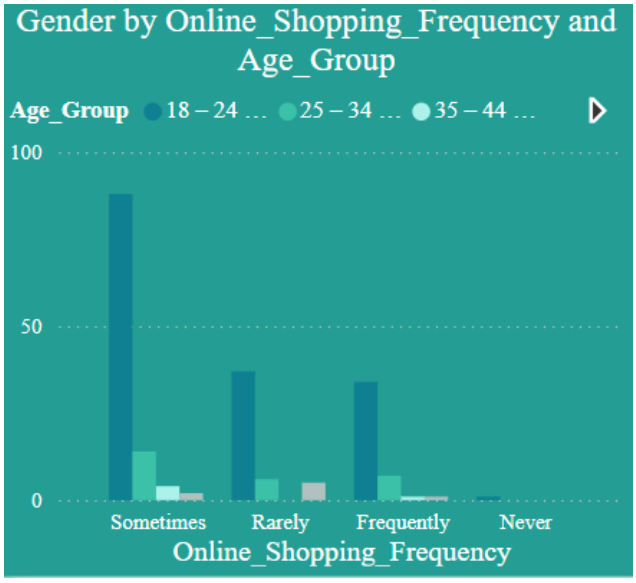
Online Shopping Frequency



Insight:

Most customers shop online occasionally ("Sometimes"), indicating moderate engagement with online shopping platforms.

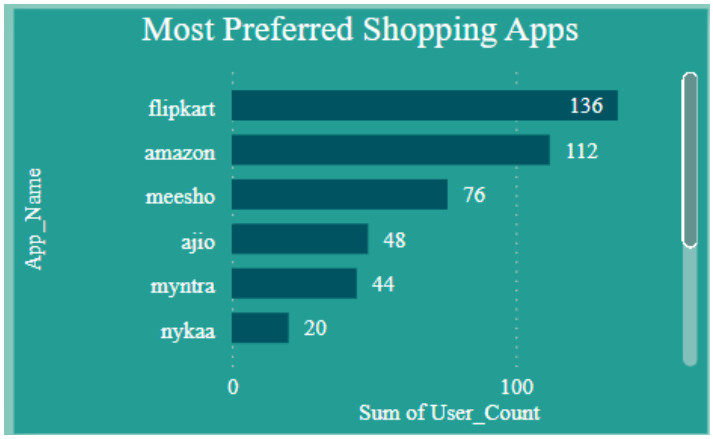
Online Shopping Frequency by Age Group and Gender



Insight:

Customers in the 18–24 age group show the highest online shopping activity, especially in the “Sometimes” category, across both genders.

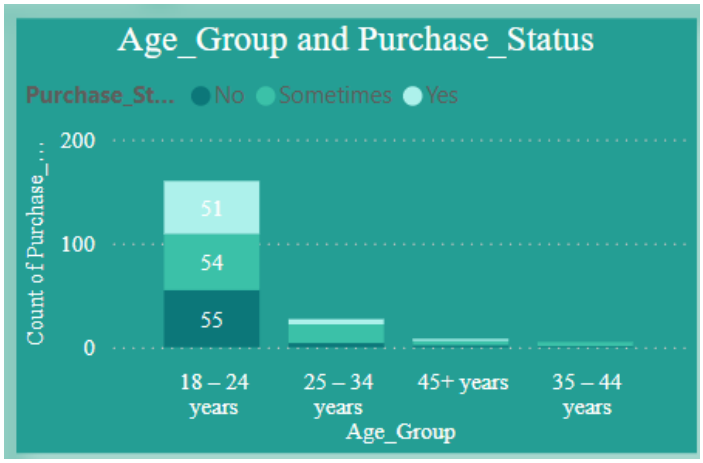
Most Preferred Shopping Applications



Insight:

Flipkart and Amazon dominate customer preference, showing strong brand trust among users.

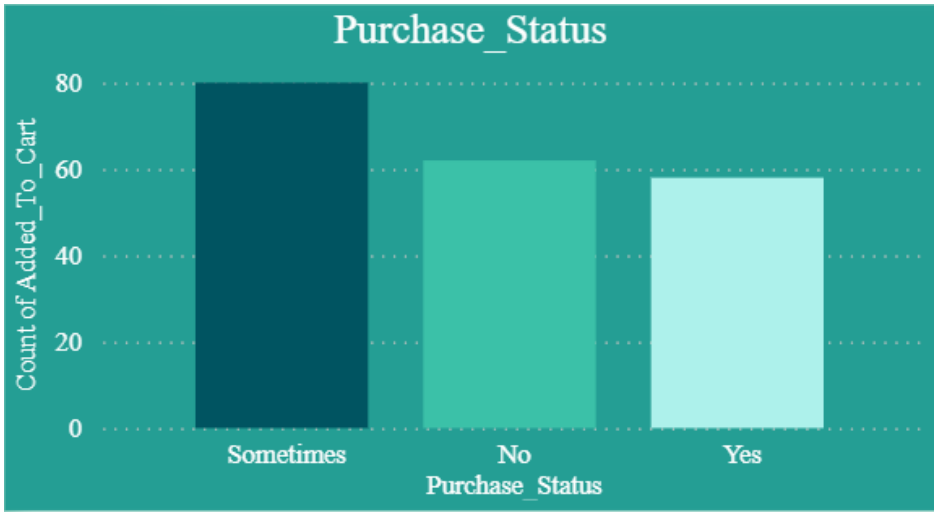
Age Group and Purchase Status



Insight:

The 18–24 age group shows the highest purchase activity, but also contributes significantly to cart abandonment.

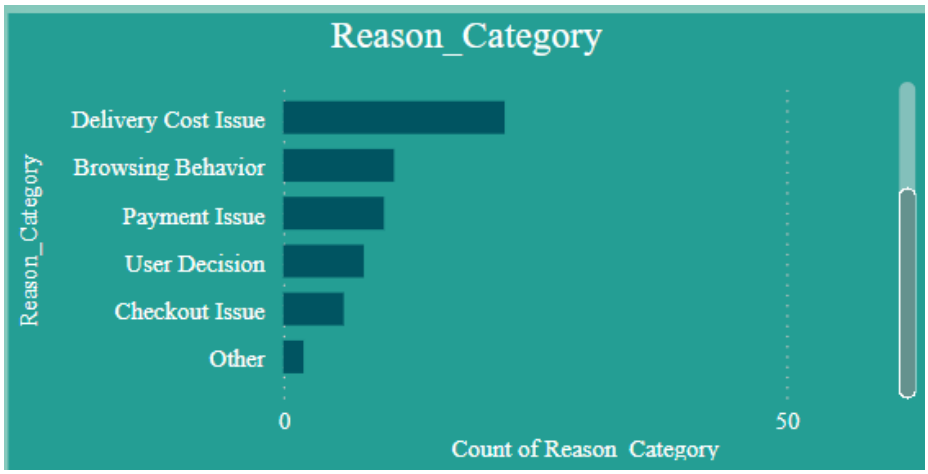
Purchase Status Distribution



Insight:

The purchase status chart shows that many customers sometimes abandon their carts before completing the purchase. This indicates that cart abandonment is a common behavior among online shoppers.

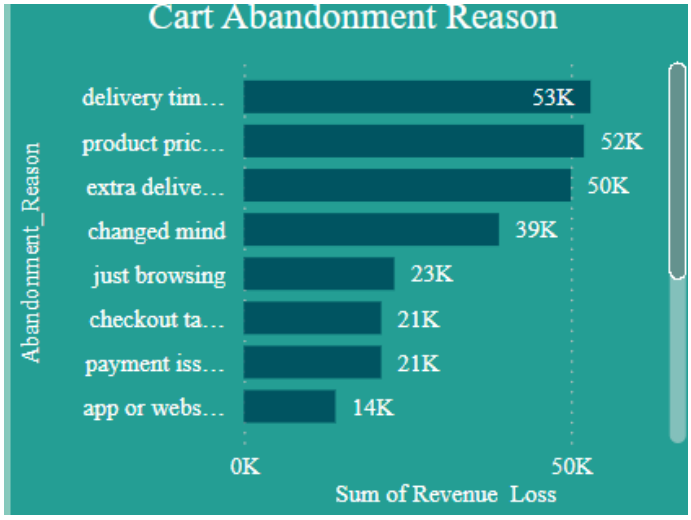
Cart Abandonment Reasons



Insight:

The main reasons for cart abandonment are long delivery time, high product price, and extra delivery charges.

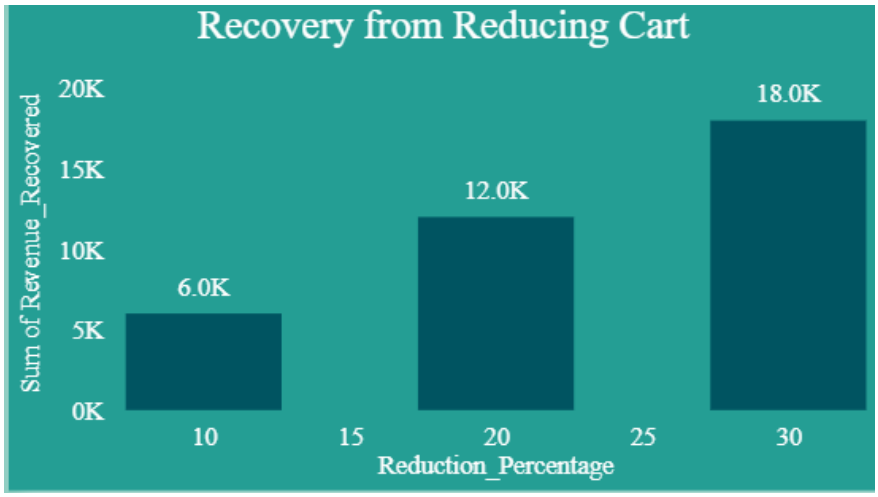
Revenue Loss Analysis



Insight:

Cart abandonment leads to significant potential revenue loss for e-commerce businesses. The analysis shows that factors such as high prices and delivery-related issues contribute to major financial losses.

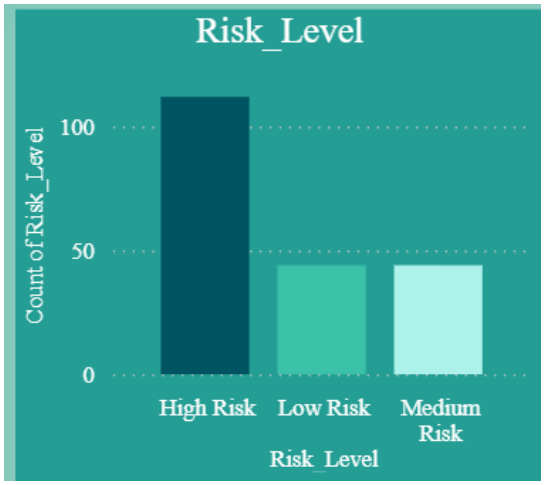
Revenue Recovery from Reducing Cart Abandonment



Insight:

Reducing cart abandonment by 10–30% can significantly increase revenue recovery

Risk Level Distribution



Insight:

Customers are classified into **High Risk, Medium Risk, and Low Risk groups** based on their purchase behavior. The analysis shows that a large number of customers fall under the **High Risk category**, indicating a higher probability of cart abandonment.

Final Dashboard

5 CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project focused on analyzing customer behavior in online shopping platforms and identifying the key factors responsible for cart abandonment. With the rapid growth of e-commerce, understanding customer purchase behavior has become essential for improving business performance and customer satisfaction.

The dataset used in this project was collected through a Google Form survey and included important attributes such as customer demographics, shopping frequency, preferred platforms, purchase status, and reasons for cart abandonment. The dataset was cleaned and processed using Python libraries such as Pandas and NumPy.

Through Exploratory Data Analysis (EDA), several patterns and trends were identified. The analysis revealed that younger customers, particularly those in the 18–24 age group, are more active in online shopping. It also showed that customers often abandon their carts due to factors such as high product prices, additional delivery charges, long delivery times, and issues during the checkout process.

A rule-based **risk score classification** was implemented to categorize customers into High Risk, Medium Risk, and Low Risk groups. This helped in identifying customers who are more likely to abandon their carts. In addition, a **revenue loss estimation** was performed to understand the financial impact of cart abandonment on e-commerce businesses.

The results were effectively visualized using Power BI dashboards, which provided clear and interactive insights into customer behavior and shopping patterns. These visualizations help businesses easily interpret the data and make informed decisions.

Overall, the project successfully achieved its objectives by analyzing customer behavior, identifying key issues related to cart abandonment, and providing meaningful insights. The findings of this project can help e-commerce platforms improve customer experience, reduce cart abandonment rates, and increase sales.

5.2 Future Work

The current project focuses on analyzing customer behavior and identifying the major factors that contribute to cart abandonment. While the analysis provides valuable insights, the system can be further enhanced by implementing advanced techniques and personalized solutions to improve customer experience and reduce abandonment rates.

One of the major improvements in future work is the implementation of a **personalized recommendation system**, where solutions are provided to customers based on their individual behavior and needs. Instead of applying the same strategy to all users, e-commerce platforms can adopt a **customer-centric approach** by analyzing shopping patterns, purchase history, and cart behavior.

Customers can be classified into different categories based on their behavior, and specific solutions can be suggested accordingly:

- **High Risk Customers** (frequently abandon carts):
  These customers can be targeted with **personalized discounts, free delivery offers, and reminder notifications** to encourage purchase completion.
- **Medium Risk Customers** (occasionally abandon carts):
  These customers can be influenced using **limited-time offers, product recommendations, and simplified checkout processes**.
- **Low Risk Customers** (regular buyers):
  These customers can be retained through **loyalty programs, reward points, and exclusive deals** to maintain engagement.

In addition to risk-based segmentation, solutions can also be customized based on specific customer needs:

- Price-sensitive customers → **Offer discounts and promotional deals**
- Delivery-sensitive customers → **Provide faster or same-day delivery options**
- Payment-related issues → **Enable multiple secure payment methods**

Another important enhancement is the development of a **real-time recommendation system** that can provide instant suggestions to customers while they are shopping. For example, if a

customer is about to abandon the cart, the system can immediately offer a discount or free delivery to retain the customer.

The project can also be extended by integrating **machine learning algorithms** to predict customer behavior more accurately. Predictive models can be used to identify customers who are likely to abandon their carts and automatically trigger appropriate actions.

Further improvements may include:

- Integration with real-time e-commerce transaction data
- Analysis of customer reviews and feedback
- Tracking user browsing behavior
- Studying seasonal trends and product categories

Additionally, the system can be developed into a **web or mobile-based application dashboard** that provides real-time insights and recommendations for business users.